

→ Theoretical Distribution

↓
 Study → Mathematical Model which is used to study the distribution of population with certain probability rule or axiom.

① Discrete Probability Distribution - Unit 3

- ① Binomial Distribution
- ② Poisson Distribution
- ③ Negative Binomial Distribution
- ④ Hyper Geometric Distribution

② Continuous Probability Distribution - Unit 4

✓ i) Normal Distribution

↳ (Gaussian Distribution)

- ii) Uniform Distribution (Rectangular Distribution)
- iii) Gamma Distribution
- iv) Beta Distribution

✓ v) Exponential Distribution

→ Binomial Distribution

→ Distribution Discrete probability Distribution.

→ Developed by James Bernoulli.

Bernoulli Trial : An experiment having only two

mutually exclusive outcomes such as : Head or

Tail, Boy or Girl, Defective or non-defective items,

Yes or No, OR Success or failure.

only on bernoulli trial → Bernoulli distribution BUT
 n no. of bernoulli trial → Binomial Distribution.

Condition under which Binomial distribution is applicable.

(i) No. of trials, n should be fixed and finite. (Sample size)

(ii) Each trial has only two mutually exclusive outcomes i.e., Success and failure.

Note: (1) Success $\rightarrow$ Required outcome
 $\hookrightarrow$ Occurrence of an event (required)
(2) Failure $\rightarrow$ Not required outcome
 $\hookrightarrow$ Non-occurrence of an event (Required).

(iii) The probability of success is denoted by p and remains constant for each trial. Similarly, the probability of failure is denoted by q and remains constant for each trial.

Here, we have:

$$p + q = 1$$

$$\Rightarrow q = 1 - p$$

var No. of
heads (0 to n)

(iv) Every trials are independent to each other.

eg, Consider a random experiment consisting of tossing a coin 10 times. $\therefore$ R.V. $= X$ (No. of Success)

(1) Here, no. of trials, $n = 10 \Rightarrow$ fixed & finite

Let, occurrence of head = success
" " " " tail = failure

and

(2) Each toss (trial) has two mutually exclusive outcomes i.e., head or tail.

③ Probability of getting head (success), $P = \frac{1}{2}$ for

each to 55

" " risk (failure): $q = \frac{1}{2}$ for

each tags

④ Every tosses are independent.

∴ The given random experiment is called
Binomial experiment.

Probability Mass Function of Binomial Distribution

Let X be a discrete R.V. that represents the no. of success in n trials.

Then, X can take the different possible values
 $0, 1, 2, \dots, n$.

Ans, the probability of getting exactly x success in n trial is calculated by using the following probability mass function -

$$P(X=x) = {}^n C_x \cdot p^x \cdot q^{n-x} \quad ; \quad x = 0, 1, 2, \dots, n$$

where, $n = \text{no. of trials (fixed)} \rightarrow n \leq 20$

$P = \text{prob. of success (constant)} \neq \text{prob. of success}$

should be > 140
0.05. ($p > 0.05$)

$q = \text{prob. of failure (1-p)}$

Note: ② The R.V. of Binomial Distribution is
~~No~~ of success

i.e., $X = \text{No of } \underline{\text{success}} \text{ (required outcome)}$

② If X follows Binomial Distribution with two parameters n and p . Then, it can be written as:

$$X \sim B(n, p)$$

③ If n and p is given, then Binomial Distribution is; completely Determined.

Q. ~~The mean~~ and If a coin is tossed 10 times then find the probabilities of getting;

- i) No heads
- ii) 2 heads
- iii) Exactly 5 heads
- iv) At least 1 head
- v) At most 2 heads
- vi) At least 9 heads
- vii) At most 8 heads

Let, $X = \text{No. of heads}$

Given

No. of tosses, $n = 10 \rightarrow n \leq 20$ ✓

Prob of getting head in each toss,

$P = \frac{1}{2}$ ~~each toss~~ $\rightarrow P > 0.05$ ✓

As both condition are satisfied we can use Binomial distribution &

$$X \sim B(n, p)$$

Then,

$$P(X=x) = {}^nC_x p^x q^{n-x} ; x = 0, 1, 2, \dots, 10$$

$$\rightarrow P(X=1) = {}^{10}C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{10-1}$$

(i) No heads;

So, $x=0$;

$\xrightarrow{\text{no. of heads}}$

$$P(X=0) = {}^{10}C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{10-0}$$

(ii) 2 heads

So, $x=2$;

$$P(X=2) = {}^{10}C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{10-2}$$

(iii) Exactly 5 heads;

$$P(X=5) = {}^{10}C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^{10-5}$$

(iv) At least 1 head;

1 or more $\rightarrow$ or just calculate every 1 or more.

$$P(X \geq 1) = 1 - P(X < 1)$$

$$= 1 - \underline{P(X=0)}$$

(v) At most 2 heads.

$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$= {}^{10}C_0 P(X=0) + {}^{10}C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{10-1} + P(X=2)$$

(vi) At least 9 heads.
9 or more.

$$\begin{aligned}
 P(X \geq 9) &= P(X=9) + P(X=10) \\
 &= {}^{10}C_9 \left(\frac{1}{2}\right)^9 \cdot \left(\frac{1}{2}\right)^{10-9} + \\
 &\quad {}^{10}C_{10} \left(\frac{1}{2}\right)^{10} \left(\frac{1}{2}\right)^{10-10}
 \end{aligned}$$

(vii) At most 8 heads

$$\begin{aligned}
 P(X \leq 8) &= 1 - P(X > 8) \\
 &= 1 - \{P(X \geq 9)\} \\
 &=
 \end{aligned}$$

Properties of Binomial Distribution

(i) It is a discrete probability distribution with two parameters n and p .

(ii) Mean of Binomial Distribution is np .
i.e., mean, $E(X) = np$.

(iii) Variance of Binomial Distribution is npq or $np(1-p)$. i.e., variance, $V(X) = npq$
and standard deviation $= \sqrt{npq}$

(iv) Variance is less than mean.

(v) Binomial distribution is symmetrical if $p=q=0.5$.

(vi) Binomial distribution is very skewed if $p > 0.5$
i.e., $p > q$

(vii) Binomial distribution is truly skewed if $p < 0.5$
i.e., $p < q$ ↓
skewed to the right

Note: (1) We use Binomial distribution practically, if

(i) $n \leq 20$

(ii) $p \geq 0.05$ & probability of success is high

(2) If a random experiment with n trials is repeated N times under similar conditions. Then, the expected freq. (No. of times) of a success is calculated as:

$$\therefore \text{Expected freq.} = N \times P(X=x)$$

where, $N =$ ~~a random~~ No. of times that a random experiment is repeated.
(population size).

$$P(X=x) = \text{Prob. pmf.}$$

ex: The mean and variance of Binomial Distribution are 6 and 4 respectively. Find the probability that of at most 2 success.

Let, $X =$ No. of success.

$$\therefore X \sim B(n, p)$$

Then, we have;

Mean, $np = 6 \rightarrow (i)$

& variance, $npq = 4 \rightarrow (ii)$

~~Dividing (i) by (ii)~~ $\therefore q = \frac{4}{6} = \frac{2}{3}$

✓✓

$$P = 1 - Q = 1 - \frac{2}{3} = \frac{1}{3} (> 0.05)$$

$$n = 6 \times 3 = 18 (\leq 20)$$

Now, $\{x = 0, 1, \dots, 18\}$

$P(\text{at most 2 success})$

$$= P(x \leq 2)$$

$$= P(x=0) + P(x=1) + P(x=2)$$

$$= {}^nC_0 p^0 q^{n-0} + {}^nC_1 p^1 q^{n-1} + {}^nC_2 p^2 q^{n-2}$$

$$= {}^{18}C_0 \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^{18-0} + {}^{18}C_1 \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^{17}$$

$$+ {}^{18}C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^{16}$$

9) In a Binomial distribution with 5 independent trials, the prob. of 1 and 2 successes are 0.4096 and 0.2048 respectively. Find p, q , mean, variance and standard deviation.

Sol. Here,

Let, $x = \text{No. of successes}$

$$x \sim B(n, p)$$

$$n = 5;$$

$$P(x=x) = {}^nC_x p^x q^{n-x}; x=0, 1, 2, \dots, 5$$

$$P(x=1) = 0.4096$$

$$P(x=2) = 0.2048$$

we know

$$P(X=2) = 0.4096$$

$$\Rightarrow {}^5C_1 \cdot p q^4 = 0.4096$$

$$\Rightarrow \frac{5!}{(5-1)! \cdot 1!} p q^4 = 0.4096 \rightarrow \textcircled{i}$$

$$\& P(X=2) = 0.2048$$

$$\Rightarrow {}^5C_2 \cdot p^2 q^3 = 0.2048$$

$$\Rightarrow \frac{5!}{(5-2)! \cdot 2!} p^2 q^3 = 0.2048 \rightarrow \textcircled{ii}$$

Now, dividing $\textcircled{i}$ & $\textcircled{ii}$,

$$\frac{{}^5C_1 p q^4}{{}^5C_2 p^2 q^3} = \frac{0.4096}{0.2048}$$

$$\Rightarrow \frac{q}{p} = \frac{0.4096 \times {}^5C_2}{0.2048 \times {}^5C_1}$$

$$\Rightarrow \boxed{0.2048 \times {}^5C_1 q = 0.4096 \times {}^5C_2 p}$$

Now, $\textcircled{i}$ becomes,

$5C_1$

Q) An experiment succeeds twice as often as it fails. Find the prob. that in the next 6 trials there will be at least 4 successes.

Sol.

Let, $X = \text{No of success}$

Also, let q be the probability of failure.

then from eqn;

$$p = 2q$$

$$\text{And, } p + q = 1$$

$$\rightarrow 2q + q = 1$$

$$\rightarrow \boxed{q = \frac{1}{3}}$$

$$\therefore \boxed{p = \frac{2}{3}} (\geq 0.05)$$

Also, $n = 6; (\leq 20)$ } Binomial distrib.

$\therefore X \sim B(n, p)$. Then,

$$P(X=x) = {}^nC_x p^x q^{n-x}; x=0,1,2,3,4,5,6$$

$$\text{So, } \cancel{P(X=4)} = \frac{{}^6C_4 \left(\frac{2}{3}\right)^4 \left(\frac{1}{3}\right)^2}{}$$

$$P(\text{at least 4 success}) = P(X \geq 4)$$

$$= P(X=4) + P(X=5) +$$

$$P(X=6)$$

$$= {}^6C_4 \left(\frac{2}{3}\right)^4 \left(\frac{1}{3}\right)^2 +$$

$${}^6C_5 \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right) +$$

$${}^6C_6 \left(\frac{2}{3}\right)^6 \left(\frac{1}{3}\right)^0$$

Q) Incidence of occupational disease in an industry is such that the workers have 20% chance of suffering from it. Find the prob. that out of 6 workers, 4 or more will ~~the~~ contract the disease.

Sol:

Let, X = No. of workers suffering the disease

Given,

$$n = 6; (n \leq 20)$$

prob. that a worker is suffering from occupational disease $\Rightarrow$

$$p = 0.2 \quad (p > 0.05)$$

$$q = 1 - 0.2 = 0.8$$

$$\therefore X \sim B(n, p)$$

$$\text{Then, } P(X=x) = {}^n C_x p^x q^{n-x}; x = 0, 1, \dots, 6$$

$$= {}^6 C_x (0.2)^x (0.8)^{6-x}$$

Now,

$$P(4 \text{ or more will contract the disease}) =$$

$$P(X \geq 4)$$

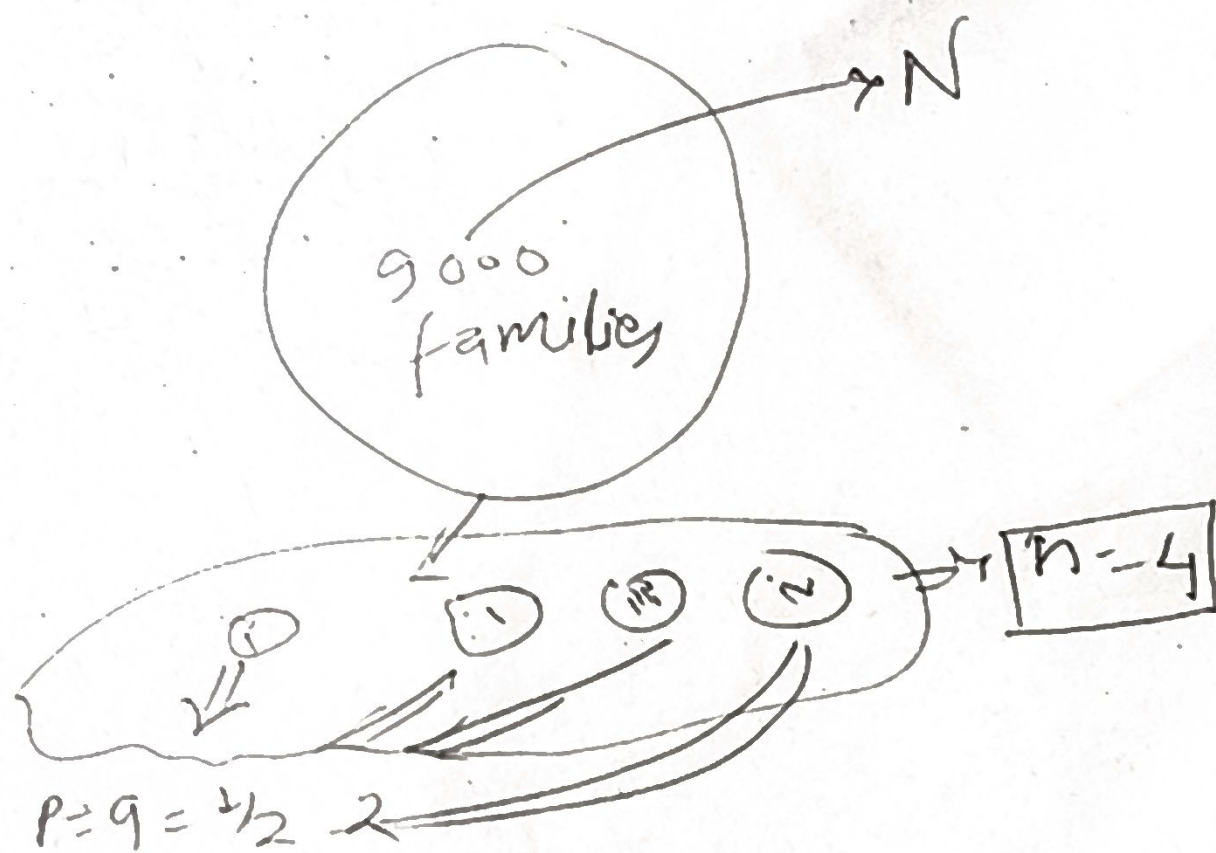
$$= P(4) + P(5) + P(6)$$

$$= {}^6 C_4 (0.2)^4 (0.8)^2 + {}^6 C_5 (0.2)^5 (0.8)$$

$$+ {}^6 C_6 (0.2)^6$$

8) out of 9000 families with 4 children each, what % would you expect to have;

- ① Two boys and two girls
- ② One boy and three girls
- ③ At least one boy
- ④ No girls (Assuming that boys and girls are equally likely)



Let, x = No. of boys.

Given, $n=4$, $N=9000$
 prob. that a child is boy;

$$\rightarrow p = 1/2$$

$$\rightarrow q = 1/2$$

$$x \sim B(n, p)$$

$$P(x=r) = {}^n C_r p^r q^{n-r} ; r = 0, 1, \dots, 4$$

$$= {}^4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{4-2}$$

(i) 2 boys and 2 girls.

no. of boys $\leftarrow$ $P(X=2) = {}^4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{4-2}$

$$= \frac{3}{8}$$

$\therefore$ % ~~part~~ of 2 boy & 2 girl is $P(X=2) = \frac{3}{8} \times 100\%$

(ii) 1 boy and 3 girls

$$P(X=1) = {}^4C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{4-1}$$

(iii) $P(X \geq 1)$

(iv) $P(X = 0)$

8) Out of 9000 families with 4 children each, how many families do you expect that consists of: no. of families

(i) Two boys and two girls

(ii) At least 1 boys

(iii) no girls

(iv) 3 boys and 1 girl

(v) At most 2 girls

(Assuming that the boys and girls are equally likely)

Let, $X = \text{No. of boys.}$

① Expected no. of families having 2 boys and 2 girls is;

$$= N \times P(X=2)$$

$$= 9000 \times \frac{3}{8}$$

② At least 1 boy = $N \times P(X \geq 1)$

$$= N \{ \cancel{P(X=0)} : 1 - P(X \leq 0) \}$$

③ No girl; = $N \times P(X=4)$

↳

④ At most 2 girls $\rightarrow 0, 1, 2$ girls
↳ 3, 2; boys

↳
↳ At least two boys;

$$\propto \underline{\underline{P(X \geq 2)}}$$

Q) 6 dice are thrown 729 times. How many times do you expect that at least 3 dice show 5 or 6.

Soln
= Let,

$X = \text{No. of dice showing 5 or 6}$

$$729 \rightarrow N$$

$$P(5 \text{ or } 6)$$

$$= \frac{1}{6} + \frac{1}{6}$$

$$= \frac{2}{6}$$

Given, $N = 729$, $n = 6$

$P =$ prob. of getting 5 or 6 on a die.

$$= \frac{2}{6}$$

$$q = \frac{4}{6}$$

$X \sim B(n, P) \Rightarrow P(X=x) = {}^n C_x P^x q^{n-x} \quad ; x=0,1,\dots,n$

Hence,

$$P(\text{at least 3 dice showing 5 or 6})$$

$$= P(X \geq 3)$$

Expected no. of or = $P(X \geq 3) \times N$

Q1) A multiple choice test contains 30 questions each with 4 answers. If a student just guesses ~~one~~ ~~quest~~ on each qns. what is the prob. that the student will ans:

- (i) exactly 7 qns. correctly
- (ii) less than 2 qns. correctly
- (iii) more than 6 qns. correctly.

Q2) The probability of a bomb hitting the target is 0.2. Two bombs are enough to destroy a bridge. If 6 bombs are aimed at the bridge, find the prob. that the bridge is destroyed.

Q3) Given,

total qns = 10 qns.

So, total options = $10 \times 4 = 40$ options

~~favorable cases~~, prob. if $X = \text{No. of}$

correct guesses then;

probability of getting correct guess =

$$\frac{\text{correct options / favorable case}}{\text{total cases}}$$

$$= \frac{10}{40}$$

$$= \frac{1}{4} = P > 0.05$$

$$\text{So, } q = 1 - P = 1 - \frac{1}{4} = \frac{3}{4}$$

$n = \text{no. of correct options} = 10 \quad (n \leq 20)$

So, $X \sim B(p, n)$.

So, by binomial distribution,

$$P(X=x) = {}^nC_x p^x q^{n-x}.$$

$$\textcircled{i} \quad P(X=7) = {}^{10}C_7 \cdot \left(\frac{1}{4}\right)^7 \cdot \left(\frac{3}{4}\right)^{10-7} = 0.003$$

$$\begin{aligned} \textcircled{ii} \quad P(X \leq 2) &= P(X=0) + P(X=1) + \\ &= {}^{10}C_0 \left(\frac{1}{4}\right)^0 \cdot \left(\frac{3}{4}\right)^{10} + \\ &\quad {}^{10}C_1 \left(\frac{1}{4}\right)^1 \cdot \left(\frac{3}{4}\right)^9 \end{aligned}$$

$$= 0.244$$

$$\textcircled{iii} \quad P(X > 6) = P(X=7) + P(X=8) + P(X=9) + P(X=10)$$

$$= {}^{10}C_7 \left(\frac{1}{4}\right)^7 \cdot \left(\frac{3}{4}\right)^{10-7} +$$

$${}^{10}C_8 \left(\frac{1}{4}\right)^8 \cdot \left(\frac{3}{4}\right)^{10-8} +$$

$${}^{10}C_9 \left(\frac{1}{4}\right)^9 \cdot \left(\frac{3}{4}\right)^{10-9} +$$

$${}^{10}C_{10} \left(\frac{1}{4}\right)^{10} \left(\frac{3}{4}\right)^0.$$

Q2) → Here,

$$n = 6 \quad (\leq 20)$$

Let, $x =$ No. of bombs hitting the target

probability of bomb hitting the target $= 0.2 = p$

$$\therefore q = 0.8$$

$$X \sim B(n, p)$$

Now

$$P(X \geq 2) = 1 - P(X < 2)$$

$$= 1 - \{P(X=0) + P(X=1)\}$$

$$= 1 - \{ {}^6C_0 \cdot (0.2)^0 \cdot (0.8)^6 +$$

$${}^6C_1 (0.2)^1 \cdot (0.8)^5 \}$$

$$= 0.34464 \frac{1}{2}$$

Poisson Distribution

↳ Another important discrete probability distribution developed by French mathematician Simeon Denis Poisson and very useful when the no. of trials (n) is very large and probability of success (p) is very small.

→ Poisson distribution is considered as a limiting case of Binomial Distribution under the following conditions:-

(i) The no. of trials (sample size), $n \rightarrow \infty$ is very large;
i.e., $n \rightarrow \infty$ ($n > 20$)

(ii) The prob. of success, p should be very small i.e., $p \rightarrow 0$ ($p < 0.05$)

(iii) $np = \lambda$ is finite

where, λ = parameter of Poisson distribution
(avg. no. of success over specified time interval or space).

→ Poisson Distribution is commonly used to study the rare-case events such as:

- (i) No. of accidents per day in a busy intersection
 - (ii) No. of wrong calls per minute
 - (iii) No. of typing mistakes in a page
- } defined under certain interval or space.

Probability Function (Formula)

Let X be a ~~R.V.~~ discrete R.V. ~~with~~ that represents the no. of success over some specified time interval or space. Then the probability of getting exactly x success is denoted by $P(X=x)$ and is obtained by using the following proof:

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}; x=0,1,2,\dots$$

where, e = Euler's const. (2.7183).
 λ = parameter (avg. no. of success over specified interval of time or space).

x = no. of success required.

Notes: (i) R.V. of Poisson distribution is no. of success. i.e., X = No. of success

(ii) If x follows Poisson distribution with parameter λ . Then it can be written as:

$$X \sim P(\lambda)$$

(iii) If the value of λ is known then, Poisson distribution is completely determined.

(iv) The avg. no. of success over specified interval of time or space, λ should be known in advance.

(v) The ~~no.~~ of avg. no. of success in an unit interval of time or space is proportional to the size of interval.

(vi) Mean of poisson distribution is λ . i.e.,

$$\text{Mean, } E(X) = \lambda$$

(vii) Variance of Poisson's distribution is λ .

$$\text{i.e., } V(X) = \lambda$$

$$\text{and standard deviation } (\sigma) = \sqrt{\lambda}$$

(viii) Mean and Variance are equal.

Q) An office switch board receives telephone calls ~~and office~~ at the rate of 3 calls per minutes on an avg. Find the probability of receiving;

(i) No calls in 1 minute interval

(ii) At least 3 calls in 1 minute interval

(iii) At most 2 calls in 5 min interval.

Soln.

Let, $X =$ No. of calls received;

Given,

avg. rate of calls^{received} per minutes (λ) = 3

(i) ~~$P(X=0) \rightarrow$ No calls per min~~

$$\therefore X \sim P(\lambda)$$

$$\text{Then, } P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, 2, \dots$$

(i) No calls per min.

$$\therefore P(X=0) = \frac{e^{-3} \cdot (3)^0}{0!} = e^{-3}$$

(ii) At least 3 call in 1 min;

$$P(X \geq 3) = \cancel{e^{-3} (3)^3 P(X=3)} + \cancel{P(X=4)} + \cancel{P(X=5)} + \dots$$
$$= 1 - P(X < 3)$$

$$= 1 - \{ P(X=0) + P(X=1) + P(X=2) \}$$

$$= 1 - \left\{ \frac{e^{-3} \cdot (3)^0}{0!} + \frac{e^{-3} \cdot (3)^1}{1!} + \frac{e^{-3} \cdot (3)^2}{2!} \right\}$$

(iii) At most 2 calls in 5 min interval

for 1 min, $\lambda = 3$

$\therefore$ for 5 min, $\lambda = 5 \times 3 = 15$

$$So, P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$
$$= \frac{e^{-15} (15)^0}{0!} + \frac{e^{-15} (15)^1}{1!} + \frac{e^{-15} (15)^2}{2!}$$

Q) Find the prob. that at most 2 defective fuse will be found in a box of 200 fuses. Previous experience shows that 2% of such fuses are defective.

Soln.

Let, $X =$ No. of defective fuses

Given,

Sample size, $n = 200$ ($n > 20$)

prob. of defective fuse, $p = 0.02$

($p < 0.05$)

So, we use Poisson distribution; $\because n$ is very large & p is very small

$$X \sim P(\lambda)$$

$$\begin{aligned} \text{where, } \lambda &= np = 200 \times 0.02 \\ &= 2 \times 2 \\ &= 4 \end{aligned}$$

Now, So, $P(X) = \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, \dots$

Now,

$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$= \frac{e^{-4} 4^0}{0!} + \frac{e^{-4} 4^1}{1!} +$$

$$\frac{e^{-4} 4^2}{2!}$$

Q) In a certain factory turning out optical lenses in a ~~certain~~ factory there is a small chance of $1/500$. For any lens to be defective. The lenses are supplied in packets of 10. What is the prob. that a packet will contain.

- (i) No defective lens
- (ii) Exactly two defective lens
- (iii) At least 1 defective lens
- (iv) At most 2 defective lens

Soln.

Let, $X =$ No. of defective lens.

$$n = 10$$

$p = 1/500$ ($p < 0.05$, small $\Rightarrow$ we use Poisson's dist.)

$$\lambda = np = \frac{1}{50}$$

Now, $X \sim P(\lambda)$.

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!} ; x = 0, 1, 2, \dots$$

$$(i) P(X=0) = \frac{e^{-\lambda} \lambda^0}{0!}$$

$$(ii) P(X=2) = \frac{e^{-\lambda} \lambda^2}{2!}$$

$$(iii) P(X \geq 1) = 1 - P(X < 1) \\ = 1 - P(X=0),$$

$$(11) P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

8) In a certain factory turning out razor blades there is a small chance of $1/500$ for any blade to be defective. Blades are supplied in packets of 10. Find the expected no. of packets containing:

(i) No defective, 1 defective, 2 defective and 3 defective respectively in contents of 10000 packets.

Soln:

Let, $x =$ No. of defective blades;

Given,

$$n = 10 \quad (n \leq 20)$$

$$N = 10000$$

$$P = 1/500 \quad (P < 0.05)$$

$P < 0.05$, so we use Poisson distribution.

~~$X \sim P(\lambda)$~~

$$\text{So, } \lambda = np = \frac{1}{500} \times 10 = 1/50$$

$$X \sim P(\lambda)$$

(i) No defectives;

$$P(X=0) =$$

$$\frac{e^{-\lambda} \lambda^x}{x!}; \lambda = 0, 1, 2, \dots$$

$$= \frac{e^{-1/50} (1/50)^0}{0!} = 0.9802$$

... expected no. of no. of times = $N \times P(X=0)$
 $= 10000 \times P(X=0)$
 $= 9801.98$
 ≈ 9802

(ii) 1 defective;

$$P(X=1) = \frac{e^{-\lambda} \lambda^1}{1!}$$

$$= \frac{e^{-1/50} (1/50)^1}{1!}$$

expected no. = $N \times P(X=1)$

(iii) Similar for 2 & 3 defectives;

Q) An air craft has 116 seats. That airlines has found from long experience that on avg. 2.5% of the people have bought the tickets for the flight do not arrive for the flight. The airlines sell 120 tickets for a flight then calculate the prob. that

(i) More than 228 people arrive for the flight
 empty seats on the flight.

(ii) There are

Let $X =$ No. of people not arriving for the flight

Given,

$$n = 120$$

$$p = 0.025$$

Since, n is large and p is small & np is small.

Then, we use Poisson distribution;

$$\lambda = np$$

$$= 120 \times 0.025$$

$$= 3$$

avg. no. of people not arriving for the flight.

$$X \sim P(\lambda)$$

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}; x=0, 1, 2, \dots$$

(i) More than 118 people arrive

$\Downarrow$
119 & 120 arrive.

So, Not arriving $\Rightarrow (120 - 119), (120 - 120)$

$\Rightarrow (1, 0)$

At most 2 people not arrive

$$\therefore P(X \leq 1) = P(X=0) + P(X=1)$$

(ii) There are empty seats. empty seats in the flight;

$\hookrightarrow$ less than 116 people arrive

$\hookrightarrow 115, 114, \dots$

So, 5, 6, $\dots$ not arrive

i.e., At least 5 don't arrive.

$$\text{So, } P(X \geq 5) = 1 - P(X \leq 4) = 1 - \{P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4)\}$$

Negative Binomial Distribution

Binomial Distribution

- (i) n is fixed
- (ii) p is same for each trial
- (iii) R.V. $\Rightarrow X = \text{No. of success}$

-ve Binomial dist.

- (i) $n \Rightarrow$ R.V.
- (ii) No. of success is fixed
or
R.V. $\Rightarrow$ No. of failures before getting the success
- (iii) p is same for each trial.

Probability function of NB (Formula)

$$(1) \quad P(X=x) = \binom{x-1}{r-1} p^r q^{x-r}; \quad x = r, r+1, \dots$$

where, $x = \text{No. of trials to get } r^{\text{th}} \text{ success}$
 $r = \text{No. of success (fixed)}$
 $p = \text{prob. of success (const. for each trial)}$

$q = 1 - p$
 $x = \text{No. of trials required.}$

$$(2) \quad P(X=x) = \binom{x+r-1}{r-1} p^r q^x; \quad x = 0, 1, 2, \dots$$

$x = \text{No. of failures before getting } r^{\text{th}} \text{ success}$

$r = \text{no. of success (fixed)}$

$p = \text{prob. of success (const.)}$

$q = 1 - p$

$x = \text{No. of failures required}$

Note: (i) The parameters of NBD are r and p .
That r and p are given.

(ii) If X follows NBD with parameters r and p .
Then, it can be written as;

$$X \sim NBD(r, p)$$

(iii) Mean = $\frac{rq}{p}$

(iv) Variance = $\frac{rq}{p^2}$

(v) Mean is less than Variance.
i.e., Variance $>$ Mean

eg: An item is produced in large no. the machine is known to produce 5% defective. A quality control inspector is examining the items by taking them at random. What is the prob. that at least 4 items are to be examined in order to get 2 defectives. Also find mean and variance.

Let, R.V.; $X =$ No. of items to be examined

Given:

No. of defective items $\leftarrow r = 2$

prob. of defective items $\leftarrow p = 0.05$

$$q = 0.95$$

Here, $X \sim \text{NBD}(r, p)$. Then

$$P(X=x) = \binom{x-1}{r-1} p^r q^{x-r}; x = r, r+1, \dots = 2, 3, \dots$$

→ A 4 items to be examined.

$$\begin{aligned} P(X \geq 4) &= 1 - P(X < 4) \\ &= 1 - \{P(X=2) + P(X=3)\} \\ &= 1 - \left\{ \binom{2-1}{2-1} (0.05)^2 \cdot (0.95)^{2-2} \right. \\ &\quad \left. + \binom{3-1}{2-1} (0.05)^3 \cdot (0.95)^{3-2} \right\} \\ &= \dots \end{aligned}$$

→ Mean = $\frac{rq}{p}$ Variance = $\frac{rq}{p^2}$

Q) A student has taken a 5 multiple choice
only. He continues to answer qns. until
he gets 3 correct answers. What is the
prob. that he gets them in at most 7
questions if he guesses at each answer.

Sol.
Let, $x =$ No of qns. to be answered
Given, $n = 5$

$$p = \frac{1}{5};$$

$$q = \frac{4}{5}$$

Here, $X \sim \text{NBD}(r, p)$, then;

$$P(X=x) = \binom{x-1}{r-1} p^r q^{x-r}; \quad x = 3, 4, \dots$$

~~3~~

① Now,

$$P(X \leq 7) = \cancel{P(X=0) + P(X=1) + P(X=2)} \\ + P(X=3) + P(X=5) + P(X=6) \\ + P(X=7) + P(X=4)$$

Q) An oil company conducts a geological study that indicates an exploratory oil well should have a 20% chance of striking the well.

① What is the prob. that 1 strike comes on 3rd well drilled

② What is the prob. that 3rd strike comes on the 7th well drilled.

Soln

Let, $X =$ No. of well drilled.

$$p = 0.2$$

$$q = 0.8$$

$\therefore X \sim \text{NBD}(r, p)$. Then,

$$P(X=x) = \binom{x-1}{r-1} p^r q^{x-r}$$

① 2 strike comes in 3rd well drills.

$$x=2$$

$$P(X=2) = \binom{3-2}{1-1} (0.2)^1 (0.8)^{3-2}$$

②

$$x=3$$

$$P(X=3) = \binom{3-3}{3-1} (0.2)^3 (0.8)^{3-3}$$

Hypergeometric Distribution:

Binomial Distribution:

- ① $n \rightarrow$ fixed
- ② R.V $\rightarrow$ No. of success
- ③ $P \rightarrow$ const. for each trial

Binomial Experiment:

- ① for finite population
- ② for infinite popⁿ

4 for this

No. of items in popⁿ, $N=11$

popⁿ size

5R

6W

popⁿ

A marble is drawn

Red or white (failure)
(Success)

1st draw: $P = 5/11$

2nd draw: $P = 5/11$

const. $\leftarrow$ same

draw with replacement

Hypergeometric distⁿ

- ① $n \rightarrow$ fixed
- ② R.V $\rightarrow$ No. of success
- ③ prob. of success $\rightarrow$ vary from trial to trial

Hypergeometric Experiment:

- ① Finite popⁿ

An experiment is performed without replacement.

Prob. function (formula):

$$P(X=x) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}$$

eg: $P(2R) = \frac{5C_2 \times 6C_1}{11C_3}$

if 3 are drawn

$$x = 0, 1, 2, \dots, \min(n, M)$$

$M \rightarrow$ No. of success

Note: (i) The parameters of Hypergeometric distribution are N, M , and n .

(ii) If X follows Hypergeometric distribution with parameter N, M , and n . Then, it can be written as;

$$X \sim HG(N, M, n).$$

(iii) Mean = $n \cdot \frac{M}{N}$

(iv) Variance = $\frac{nM(N-M)(N-n)}{N^2(N-1)}$

N = popⁿ size

M = No. of success in popⁿ.

n = sample size

x = No. of success req^d.

Q7 A box contains 20 bulbs among them 6 are defectives. If 5 bulbs are drawn at random without replacement. What is the prob. that

(i) Exactly 4 bulbs are defective

(ii) At least 2 bulbs are defective

(iii) No defective.

Soln

Let, x = No. of defectives

Given,

Popⁿ size (N) = 20

No. of sample size (n) = 5

No of success (M) = 6

$\therefore X \sim HG(N, M, n)$; ~~min of~~
 ~~$x \geq 0, n \leq N$~~
min. of (n, M)